Ontology-Grounded Agentic AI

**THE WORLD'S
ONTOLOGY ECOSYSTEM**
UNIVERSITY AT BUFFALO and NCOR

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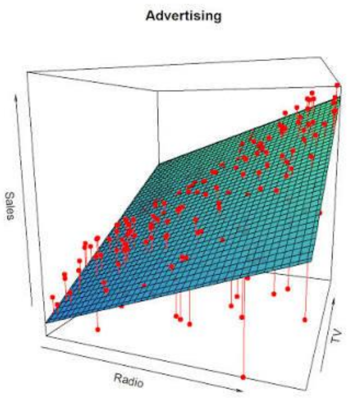
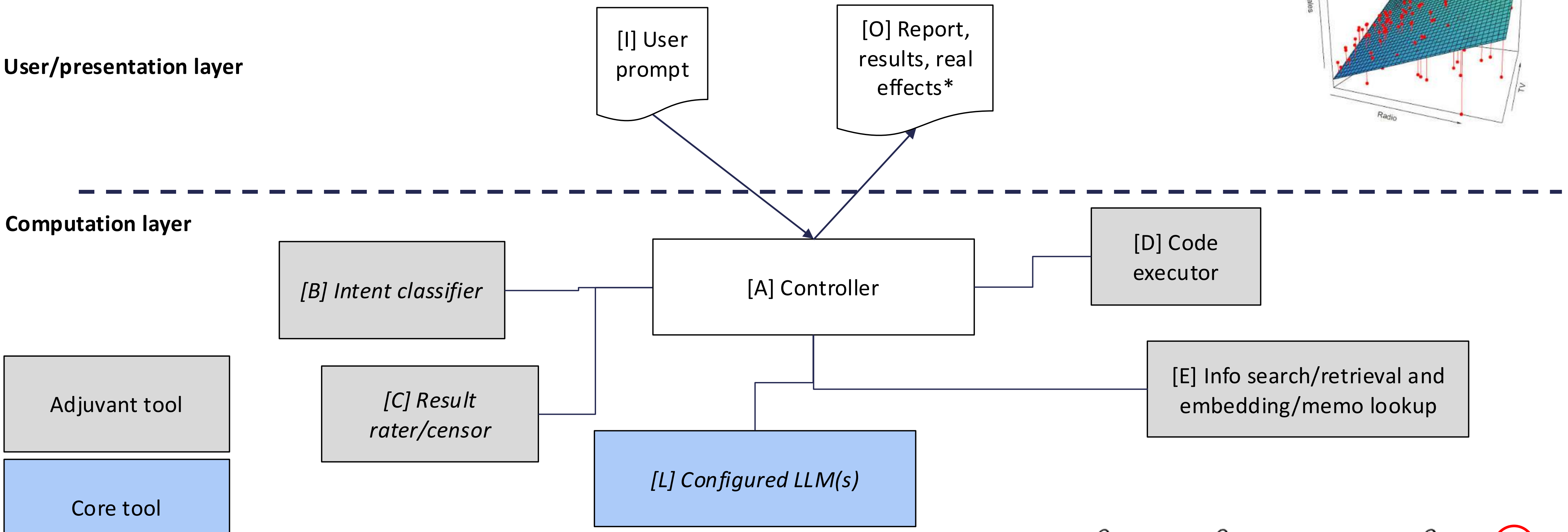
Agentic AI is a classical LLM AI wrapper architecture to enable a higher quality of results



What agentic AI is



Typical simple sequences:

I- A – B - A– E –A – L – A - C –A - O

I -A – B – A – E – A – L – A- C-A – D – A - O

$$y_t = x_{t1}\beta_1 + x_{t2}\beta_2 + \ldots + x_{tK}\beta_K + \varepsilon_t$$

* e.g. hacking into a system

Since agentic AI works with stochastic core components, errors and hallucinations cannot be avoided

Properties of agentic AI

- No intentions or goals
- No understanding of text and context
- Poor artificial (transformer-based) context window [see right-hand side]
- Deterministic controller and some deterministic adjuvant components, but stochastic main components
- Therefore, errors and hallucinations by necessity. *Not suitable for process automation whenever reliable transactions needed* (e.g. money transfer, killing of humans or object destruction in warfare)
- **No autonomous system**

- Norman Paulsen[(1)] set up an experiment in which he gave LLMs simple factual sentences and asked for information extraction (e.g. count the number of objects of a certain colour in input sentences). The models' performance collapses when the context exceeds 200 tokens (syllables), but every 7-8 year old child can perform this task.
- Binyan Xu et al.[(2)] show that memo lookup based agentic AI cannot generalise to novel situations no matter how much information gets stored; this is also true if weights get updated[(3)]

1) Paulsen, N. (2025). Context is what you need: The maximum effective context window for real world limits of llms. arXiv preprint arXiv:2509.2136
2) Xu, B., Dai, X., & Zhang, K. (2026). Contextual agentic memory is a memo, not true memory. arXiv preprint arXiv:2604.27707.
3) Mohsin, M. A., et al. (2025). On the fundamental limits of LLMs at scale. arXiv preprint arXiv:2511.12869.

The LLM configuration guarantees errors and hallucinations; the LLM does not think at all

Hallucination overview

$$y_t = x_{t1}\beta_1 + x_{t2}\beta_2 + \ldots + x_{tK}\beta_K + \varepsilon_t$$

Training data always incomplete by necessity

Refinement massively biased for non-empty and answers aligned with postmodern collectivism (wokeness)

Simulated data lead to sub-parity with humans

Configuration

- Basic unsupervised training
- Supervised refinement
- Reinforcement learning using policy optimisation (bulk parametrisation)

Intent classification

Web search/info retrieval

Output generation

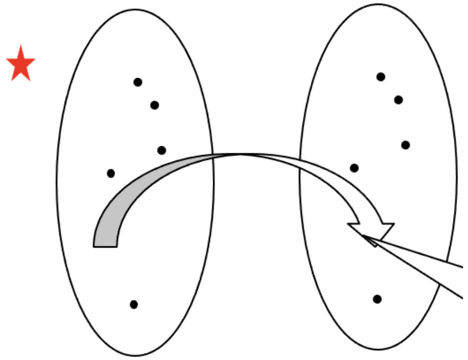
Post generation checking

Prompt contextualisation often fails

Errors in info retrieval a necessity

LLM cannot know when to halt
- Therefore, it cannot generate exact sequence
- Can produce any error type

Merely based on stochastic ensembles or other stochastic components



All the above assertions can be proven by reduction to Turing's Halting problem

Source: Banerjee 2024. arXiv:2409.05746v1

Because of the unreliability of agentic AI, there is now a trend to combine agentic AI with ontologies, a form of symbolic AI

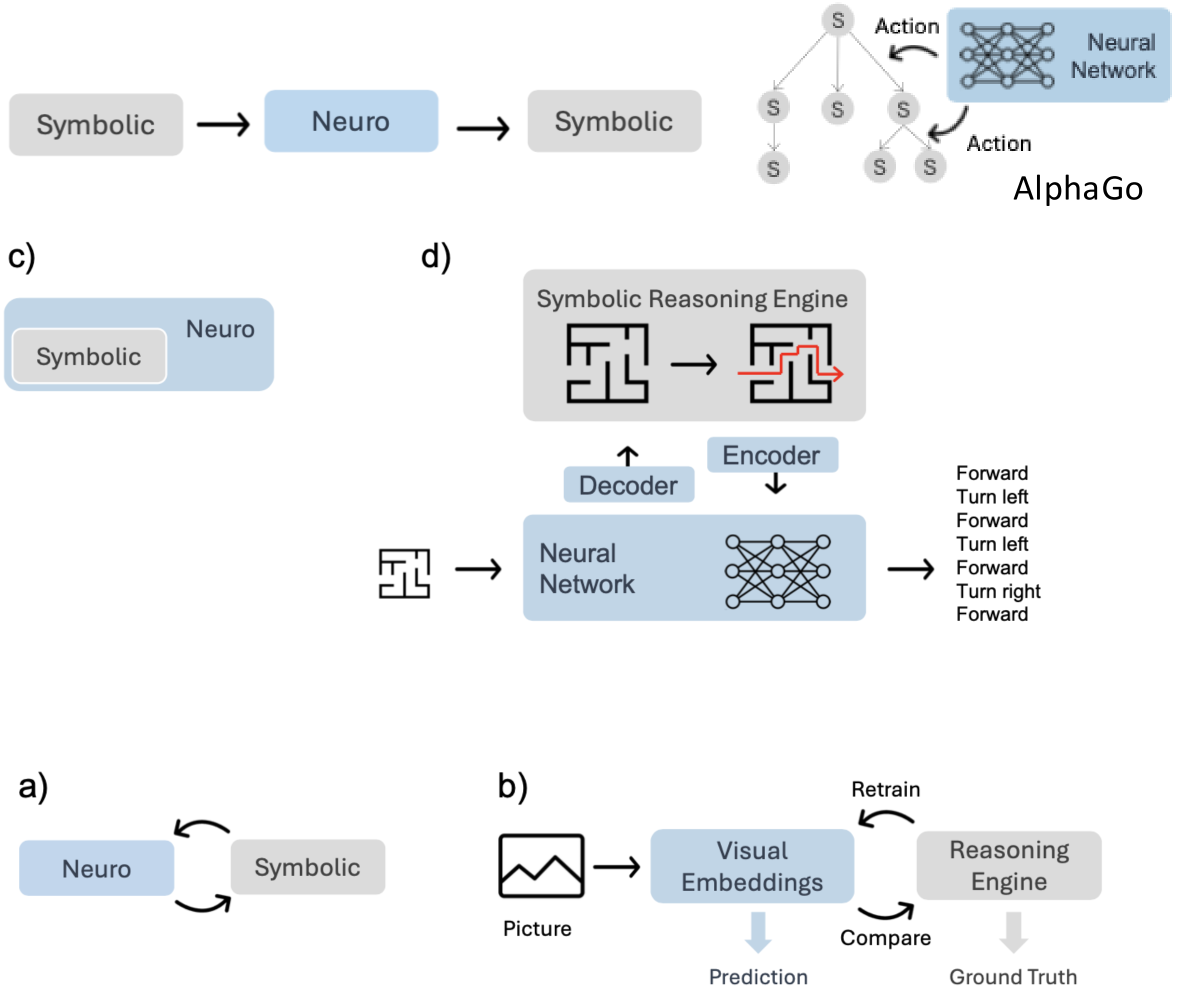
Goals of combining stochastic and symbolic AI

1. Less configuration data need due to prior knowledge - feasible
2. Certifiable AI - feasible [1]
3. "Explainable AI" - to some extent feasible
4. Better reliability, quality and control - feasible
5. Machine intelligence - **impossible**

[1] Landgrebe Certifiable AI 2022 https://doi.org/10.3390/app12031050

There are different approaches to combining ontologies and AI

Possibilities of combining stochastic and symbolic AI



Sequential (only approach that can guarantee results)

Nested

$$y = g_{\text{symbolic}}(x, f_{\text{neural}}(z))$$

Example: AlphaGo; x: symbolic context, z - input

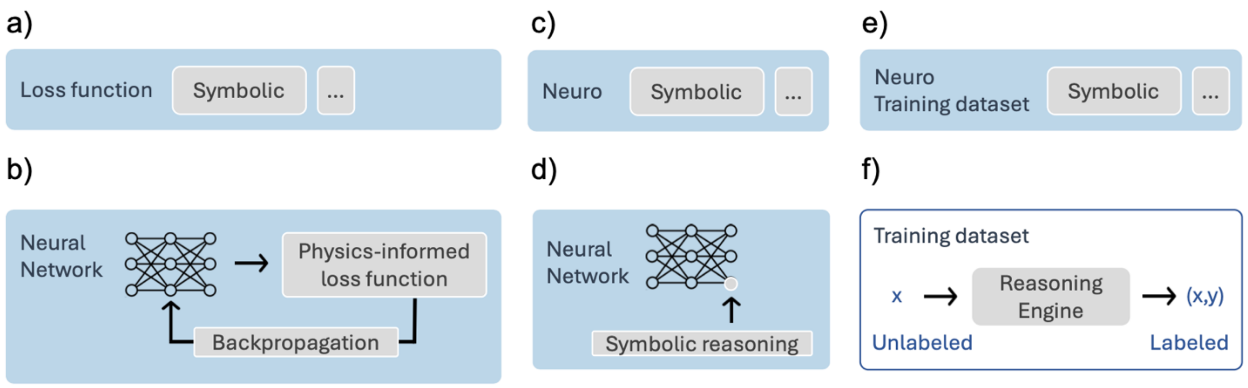
Cooperative

$$z^{(t+1)} = f_{\text{neural}}(x, y^{(t)}), \quad y^{(t+1)} = g_{\text{symbolic}}(z^{(t+1)}),$$

Source: Bougzime, …, Demoly: Unlocking the Potential of Generative AI through Neuro-Symbolic Architectures – Benefits and Limitations arXiv:2502.11269v1

There are different approaches to combining ontologies and AI

Possibilities of combining stochastic and symbolic AI



Compiled with *symbolic*

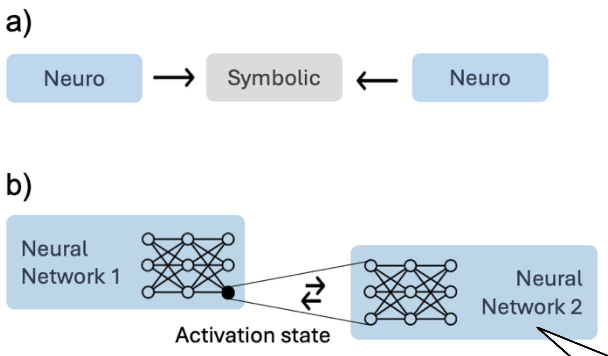
a-b) *loss*, $\mathcal{L} = \mathcal{L}_{\text{task}}(y, y_{\text{target}}) + \lambda \cdot \mathcal{L}_{\text{symbolic}}(y)$

c-d) *activation*, $y = g_{\text{symbolic}}(x)$ g: activation function at the neuron level

e-f) or *training* $\mathcal{D}_{\text{train}} = \{(x, g_{\text{symbolic}}(x)) \mid x \in \mathcal{X}\}$

g –annotation function

"Injective models provide small gains in performance relative to their unenriched counterparts. These gains tend to diminish with additional training data for non-enriched DNNs. The claim that injective models offer greater transparency than non-injective DNNs is open to question." Lappin 2025 https://qmro.qmul.ac.uk/xmlui/handle/123456789/113314 (on **practical effects of injection in language-related applications**)



Ensemble (Neuro →Symbolic ←Neuro)
uses multiple interconnected NNs via a symbolic "fibring function", which enables them to collaborate and share information while adhering to symbolic constraints

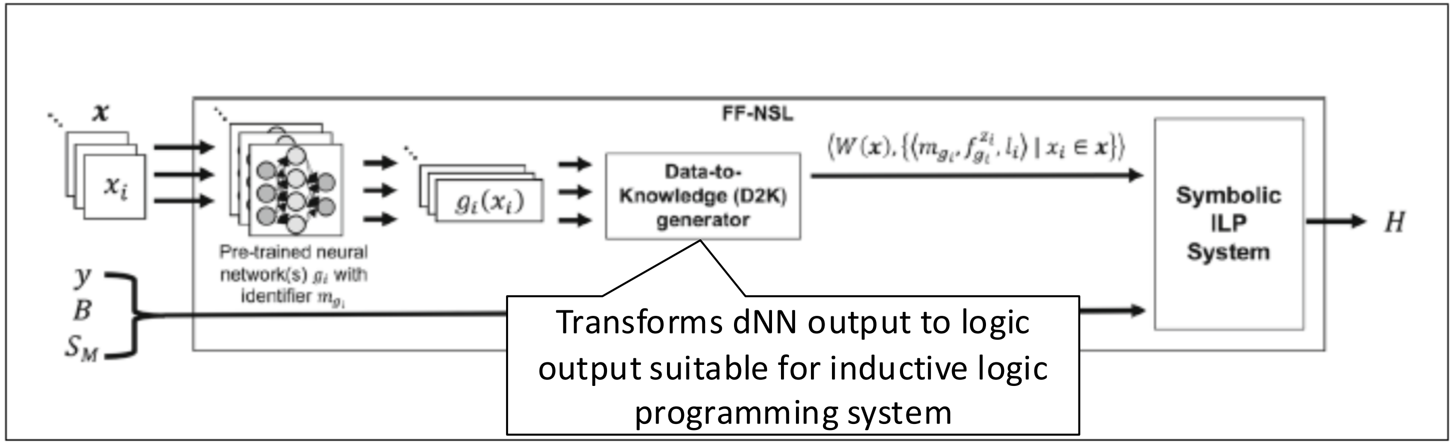
The two dNN communicate via activation states to share representations

Source: Bougzime et al. arXiv:2502.11269v1

Component federation combines stochastic and deterministic AI in a fixed or controller-determined sequence

Sequential hybrid example

$$\hat{y} = \hat{T}_0 = T_1 \circ f_1 \circ \cdots \circ f^{\theta}_{m-1} \circ T^{\kappa}_{n-1} \circ f^{\lambda}_{m} \circ T_n$$



General approach: sequential combination of stochastic and deterministic components, often with a deterministic component at the end.

Example: Image classification
Hybrid consists of a **dNN for extracting features** from images, an **interface** component that assigns labels to these features, and **a logic based** component, an Inductive Logic Program (ILP), that learns rules from these labelled features.
x: Image sequence, y, label, H: hypothesis (label explanation) that can give label to new observation (Cunnington 2023,
DOI: 10.48550/arXiv.2106.13103)

The approach by Cunnington et al. leads to a very strong performance improvement in image classification as opposed to purely stochastic methods **in a restricted problem space.**

However, with ILP, one can automatically expand propositional ontologies consisting of logical clauses under restricted conditions based on input clauses

Inductive Logic Programming

Inputs

Background Knowledge (B): Pre-existing facts and rules about the domain.

Example: `parent(ann, bob)`and `female(ann).`

Positive Examples (E^+): Specific instances that should be true.

Example: `mother(ann, bob)`

Negative Examples (E^-): Specific instances that must *not* be true.

Example: `not(mother(bob, ann)).`

Goals: The ILP system searches a space of possible logical clauses to construct a hypothesis program Hsuch that:

Completeness: The hypothesis, combined with background knowledge, entails all positive examples:

$$B \cup H \vDash E^+$$

Consistency: The hypothesis does not entail any negative examples:

$$B \cup H \nvDash E^-$$

Output:

In this family relationship example, the ILP system induces the general rule:

$$\texttt{mother}(\mathrm{X}, \mathrm{Y}) \leftarrow \texttt{parent}(\mathrm{X}, \mathrm{Y}) \land \texttt{female}(\mathrm{X})$$

An ILP system can generate ontological clauses under direct or indirect human supervision

Example generated with gemini.google.com, further reading: Nienhuys-Cheng, de Wolf: Foundations of Inductive Logic Programming (Springer, 1997); De Raedt: Logical and Relational Learning (2008)

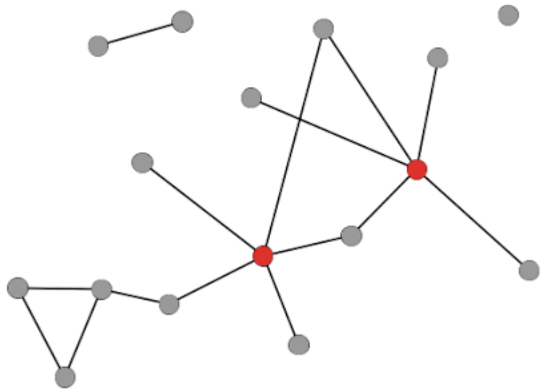
When designing hybrid systems, BFO-style ontologies can form the basis of propositional network ontologies, the workhorse to obtaining better results

Another type of ontology

- The type of ontology needed for hybrid AI is not a taxonomy with definitions, but a network of ontological propositions
- The process schema is:

LLM output → LLM-based translation into FOL → MTP validation of clauses against proposition ontology

- The mechanical theorem (MTP) prover ensures the semantics of the LLM output to guarantee performance at the expense of scope
- A proposition ontology is a network of propositions in which the nodes are entities shared by the clauses
- These nodes are taken from a taxonomy, ideally based on BFO+



The **cat** *is* on the mat
Cats *are* **carnivores**
A **carnivore** *is* an animal that eats flesh

fat: node nouns, *italic*: verb nodes

But the addition of deterministic (knowledge) components has narrow limitations

Limits of neuro-symbolic AI

1. Ontologies depend on human perception, abstraction, reasoning and intersubjective planning and **consensus** (for collaborative ontology projects, so almost all of them)
2. Therefore, world knowledge embedded in ontologies is restricted and highly context- and also perspective-dependent. Ontology scope is always **local**. This restricts usage to special problems. Ontology based solutions **do not generalise**.
3. Ontologies contain a **static** knowledge and cannot be dynamically altered (in contrast to active perception, for example). This makes them **unflexible**.
4. Ontologies can only capture **regularities**, but complex systems produce irregularities all the time and in non-predictable patterns.
5. Neuro-symbolic AI systems need to be carefully prepared and configured for the task at hand. **They are not intelligent, but useful automata.**

Ontologies improve AI models but cannot create machine intelligence.

Appendix

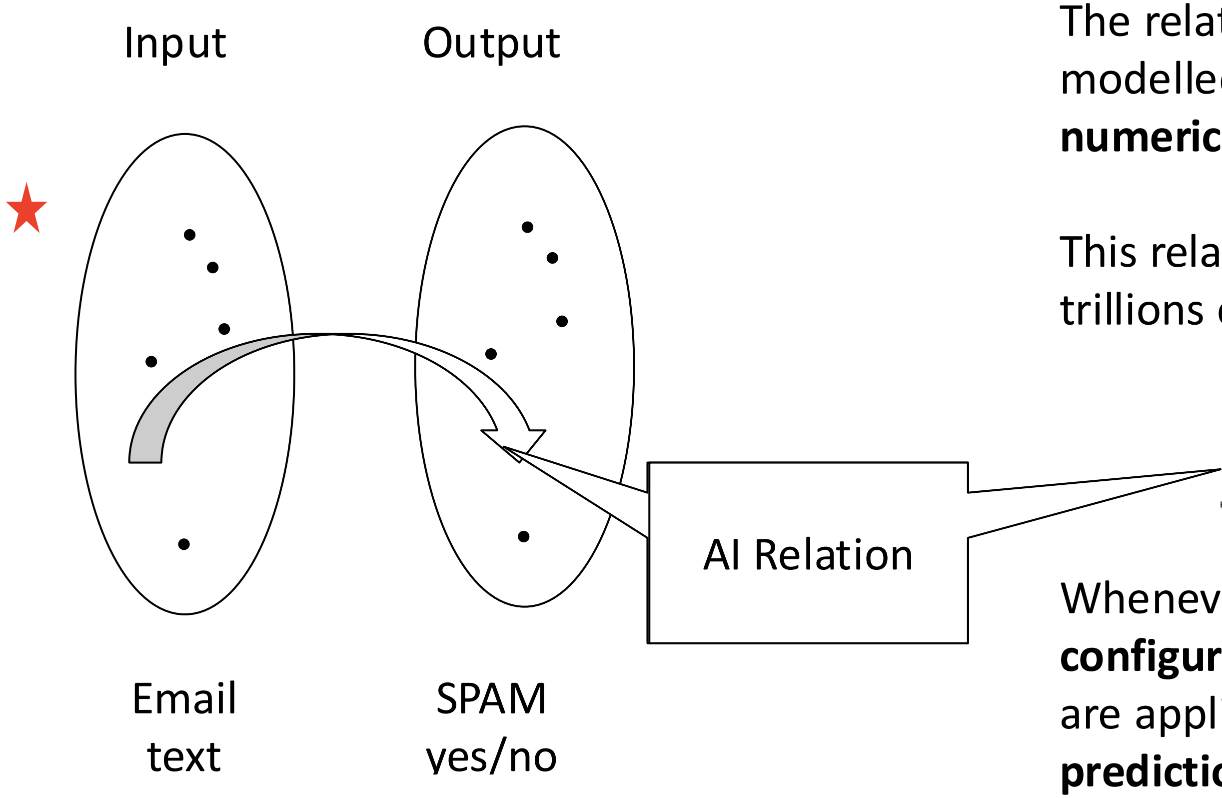
Intelligence is a dominant property of complex living systems, as human language is

Dominant properties of complex systems

- **Complex systems** also have many properties that lead to regular patterns: In humans, it is the vital processes: breathing, heartbeat, sleep, drinking, eating, digestion, sexuality. However, these can be temporarily interrupted by the erratic character of higher functions, even breathing. **Regularity**, however, **enables their mathematical modeling,** as the perfect ECG models of the healthy and diseased heart show
- Dominant characteristics of complex systems are those that **distinguish** their bearer from other systems and **have no patterns**: In humans, these are consciousness, self-awareness, feelings, decision-making, intelligence, language.
- In the case of the sun, its dominant property as a nuclear fusion emitter as opposed to its property as a gravity mass that it shares with its satellites, which is highly regular.
- **Dominant properties of complex systems can never be mathematically mapped due to the limitation of mathematics to regular patterns.**

Today, AI is mostly seen as statistical learning for automation and pattern identification purposes

Statistical (“machine”) learning



$$\langle \texttt{email}, \texttt{spam}[\texttt{0}|\texttt{1}] \rangle$$

The relationship between input and output variables is modelled using a **loss function** which is **minimised via numerical optimisation procedures**

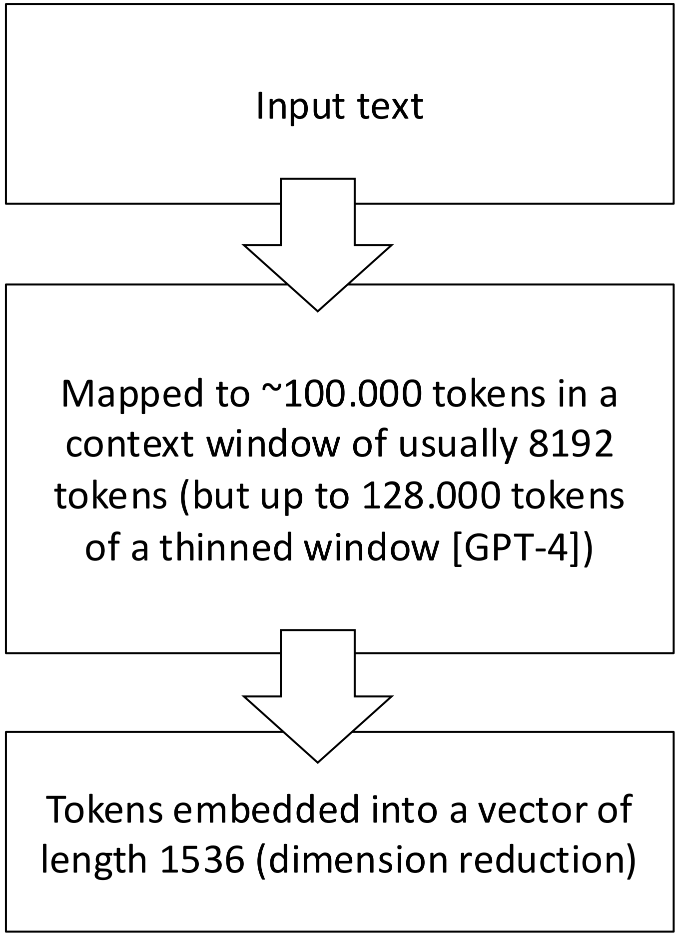
This relationship determines the parameters of the model – trillions of them in the latest LLM

$$f : \mathbb{R}^k \mapsto \mathbb{R},\ f(\boldsymbol{x}) = \hat{y}$$

Whenever a model obtains data that are **outside of the configuration distribution (** ★**)** the parameters of the model are applied to previously unseen data, **yielding large prediction errors**

LLM with high dimensional vector spaces are hopelessly underdetermined

LLM dimensions

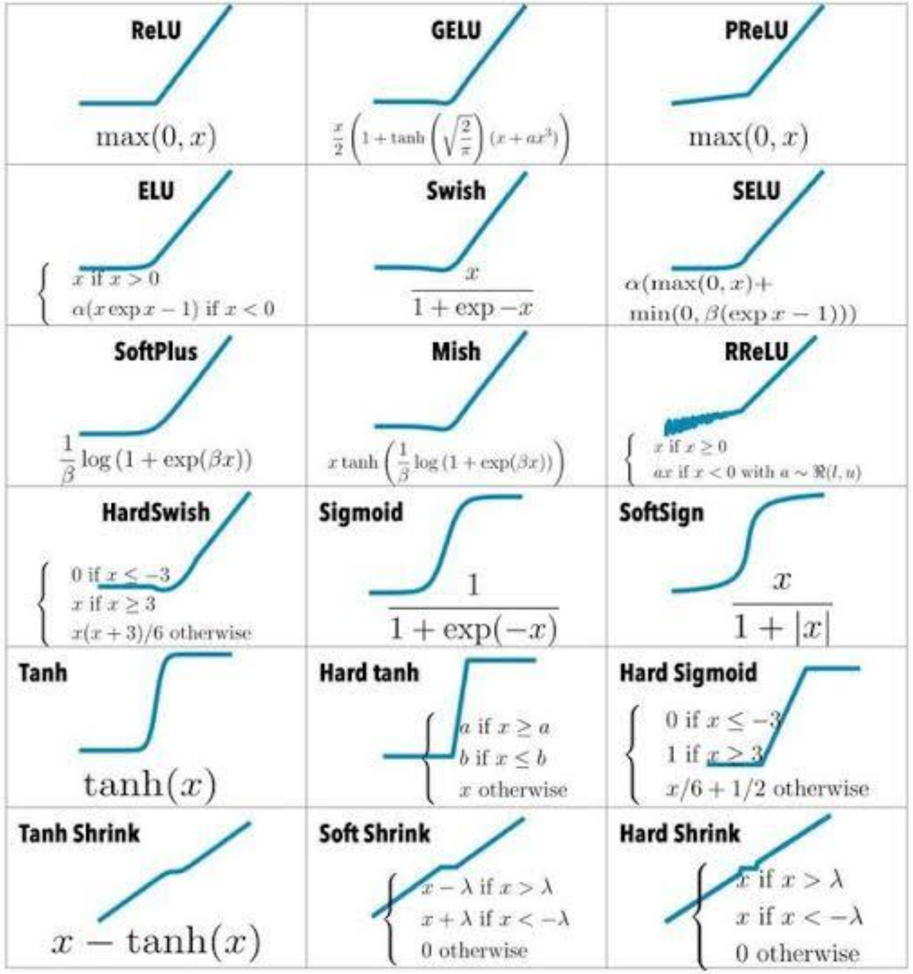


- After dimension reduction, GPT-4 uses a matrix with 1536 (3×2^9) columns
- This means that to achieve the same density obtained from 100 data points in one dimension [sufficient to test height difference between males and females, for example], one would need 100^{1536} data points. But our observable universe only has only roughly ~10^{80} protons (Eddington number N_{edd}) – thus that density cannot be achieved.
- To compensate for the curse of dimensionality, the LLM have to undergo a **massive reinforcement learning** with emulated output evaluation (Reinforcement Learning from Human Feedback, RLHF)
- Nevertheless, a typical **RLHF** has **„only" 16 million input tokens** over thousands of training steps
- Therefore, the **distribution matrix is still massively underdetermined** and creates a thinly populated sample space since for a p-dimensional unit ball with N uniformly distributed data points, the median distance from the ball's centre to the closest data point is given by

$$d(p, N) = \left(1 - \frac{1}{2}^{1/N}\right)^{1/p}$$

- So here **d(1536,16*10^7)=0,99** so that 99% of the data points are close to the boundary of the data sphere than to the centre. **This means that the LLM's regression hyperplane is massively underdetermined, especially for edge cases**. This is what produces the errors.

Neural Network Activation Functions: a small subset!



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Scientific interests and publications

Theory of AI
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